

Exercice n 5:

$$1) A = \text{mat}(f, B_1, B_2)$$

$$\text{avec } B_1 = (e_1, e_2, e_3) = \left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right)$$

$$B_2 = (u_1, u_2) = \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right)$$

$$A = \begin{pmatrix} \underset{f(e_1)}{1} & \underset{f(e_2)}{-1} & \underset{f(e_3)}{2} \\ 0 & -2 & 1 \end{pmatrix} \begin{matrix} u_1 \\ u_2 \end{matrix}$$

$$f(e_1) = f\left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = u_1$$

$$f(e_2) = f\left(\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} -1 \\ -2 \end{pmatrix} = -u_1 - 2u_2$$

$$f(e_3) = f\left(\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 1 \end{pmatrix} = 2u_1 + u_2$$

$$2- \quad * A + 5I_2 \rightarrow \text{impossible}$$

$$* {}^t A A = \begin{pmatrix} 1 & 0 \\ -1 & -2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 2 \\ 0 & -2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & -1 & 2 \\ -1 & 5 & -4 \\ 2 & -4 & 5 \end{pmatrix}$$

$$* A ({}^t A - 2I_3) \rightarrow \text{impossible}$$

$$3) S = A {}^t A - 6I_2$$

$$A {}^t A = \begin{pmatrix} 1 & -1 & 2 \\ 0 & -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -1 & -2 \\ 2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 6 & 4 \\ 4 & 5 \end{pmatrix}$$

$$S = \begin{pmatrix} 6 & 4 \\ 4 & 5 \end{pmatrix} - 6I_2$$

$$= \begin{pmatrix} 0 & 4 \\ 4 & -1 \end{pmatrix}$$

$$* S^2 = \begin{pmatrix} 0 & 4 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} 0 & 4 \\ 4 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 16 & -4 \\ -4 & 17 \end{pmatrix}$$

$$S^2 + S - 10I_2 = \begin{pmatrix} 16 & -4 \\ -4 & 17 \end{pmatrix} + \begin{pmatrix} 0 & 4 \\ 4 & -1 \end{pmatrix} - \begin{pmatrix} 10 & 0 \\ 0 & 10 \end{pmatrix}$$

$$= \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix} = 6I_2$$

$$b) S^2 + S - 10I_2 = 6I_2$$

$$S^2 + S = 16I_2$$

$$\Rightarrow S(S + I_2) = 16I_2$$

$$\Rightarrow S\left(\frac{1}{16}(S + I_2)\right) = I_2$$

done Set invet $S^{-1} = \frac{1}{16}(S + I_2)$.

$$4) a) N = {}^t A \cdot A \cdot S I_3$$

$$= \begin{pmatrix} -4 & -1 & 2 \\ -1 & 0 & -4 \\ 2 & -4 & 0 \end{pmatrix}$$

Soit $a = (x, y, z) \in \mathbb{R}_3$

On pose $X = \underset{B_1}{\text{mat}}(a) = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$$Y = \underset{B_1}{\text{mat}}(g(a)) = NX = \begin{pmatrix} -4x - y + 2z \\ x - 4z \\ 2x - 4y \end{pmatrix}$$

$$\Rightarrow g(a) = g(x, y, z)$$

$$= \begin{pmatrix} -4x - y + 2z \\ x - 4z \\ 2x - 4y \end{pmatrix}$$

$$d) M = \text{mat}(g, B'_1)$$

$$= \begin{pmatrix} g(e'_1) & g(e'_2) & g(e'_3) \\ -4 & -4 & -1 \\ 0 & 4 & 3 \\ 1 & 2 & -4 \end{pmatrix} \begin{matrix} e'_1 \\ e'_2 \\ e'_3 \end{matrix}$$

$$g(e'_1) = g(0, 1, 1) = (1, -4, -4) = e'_3 - 4e'_1$$

$$g(e'_2) = g(0, 0, 1) = (2, -4, 0)$$

$$= 2e_1 - 4e_2$$

$$= 2e'_3 - 4(e'_1 - e'_2)$$

$$= 2e'_3 - 4e'_1 + 4e'_2$$

$$+ g(e'_3) = g(e_1) = g(1, 0, 0)$$

$$= (-4, -1, 2)$$

$$= -4e_1 - e_2 + 2e_3$$

$$= -4e'_3 - e'_1 + 2e'_2 + 2e'_3$$

$$= -e'_1 + 2e'_2 - 4e'_3$$

$$c) M = \text{mat}(g, B'_1)$$

$$N = \text{mat}(g, B_1)$$

$$P = \text{Pass}(B_1, B'_1)$$

$$M = P^{-1} N P$$

$$(N = P M P^{-1})$$

$$e'_1 = e_2 + e_3 \quad ; \quad e'_2 = e_3 \quad ; \quad e'_3 = e_1$$

$$b) B'_1 = (e'_1, e'_2, e'_3)$$

Soit $\alpha, \beta, \gamma \in \mathbb{R}$

$$\text{tq } \alpha e'_1 + \beta e'_2 + \gamma e'_3 = 0$$

$$\alpha(e_2 + e_3) + \beta e_3 + \gamma e_1 = 0$$

$$\Rightarrow \gamma e_1 + \alpha e_2 + (\alpha + \beta) e_3 = 0$$

on (e_1, e_2, e_3) est l.b. de $\mathbb{R}_3 \Rightarrow$

$$\Rightarrow \boxed{\alpha = \beta = \gamma = 0}$$

alors B'_1 est une famille libre

$$\text{et } \text{card}(B'_1) = 3 = \dim \mathbb{R}^3$$

ce B'_1 est une base de $\mathbb{R}^3$

$$c) P = \text{Pass.}(B_1, B'_1)$$

$$P = \begin{pmatrix} e'_1 & e'_2 & e'_3 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} e_1 \\ e_2 \\ e_3 \end{pmatrix}$$

$$\begin{cases} e'_1 = e_2 + e_3 \\ e'_2 = e_3 \\ e'_3 = e_1 \end{cases} \Rightarrow \begin{cases} e_2 = e'_1 - e_3 \\ e_3 = e'_2 \\ e_1 = e'_3 \end{cases}$$

Ex n°6

1) $u(P) = -2P + (x+1)P'$

Soient A et $B \in \mathbb{R}_n[x]$

$$\begin{aligned} d \in \mathbb{R} \\ u(dA+B) &= -2(dA+B) + (x+1)(dA+B)' \\ &= -2dA - 2B + (x+1)dA' + (x+1)B' \\ &= d(-2A + (x+1)A') - 2B + (x+1)B' \\ &= d u(A) + u(B) \end{aligned}$$

$\Rightarrow u$ est une app linéaire

Soit $P \in \mathbb{R}_n[x] \Rightarrow \deg P \leq n$

$$\deg(P') \leq n-1$$

$$\deg((x+1)P') \leq n-1+1 = n$$

$$\Rightarrow \deg(u(P)) \leq n$$

donc $u(P) \in \mathbb{R}_n[x]$

d'où u est un endomorphisme de $\mathbb{R}_n[x]$.

2) $A = \text{mat}(u, B) =$

$u(1)$	$u(x)$	$u(x^2)$	$u(x^3)$
-2	1	0	0
0	-1	0	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$
0	0	0	0

$\begin{matrix} 1 \\ x \\ x^2 \\ \vdots \\ x^{n-1} \end{matrix}$

$$u(1) = -2$$

$$u(X) = -2X + (X+1)$$

$$= -X + 1$$

$$u(X^k) = -2X^k + (X+1)kX^{k-1}$$

$$= -2X^k + kX^k - kX^{k-1}$$

$$= (k-2)X^k - kX^{k-1}$$

$$u(X^n) = (n-2)X^n + nX^{n-1}$$

$$\exists a) P_k = X^{n-k}, \quad 0 \leq k \leq n$$

$$B = (P_k; 0 \leq k \leq n) = (X^n; X^{n-1}, \dots, X, 1)$$

$$P_0 = X^n$$

$$P_1 = X^{n-1}$$

$$P_{n-1} = X; P_n = 1$$

or $(1, X, \dots, X^n)$ est la RC de $\mathbb{R}_n[X]$.

-donc cette famille est libre.

Soient $\alpha_0, \dots, \alpha_n \in \mathbb{R} / \sum_{i=0}^n \alpha_i P_i = 0$.

$$\Leftrightarrow \alpha_0 P_0 + \alpha_1 P_1 + \dots + \alpha_n P_n = 0$$

$$\Leftrightarrow \alpha_0 X^n + \alpha_1 X^{n-1} + \dots + \alpha_{n-1} X + \alpha_n = 0$$

$$\Leftrightarrow \alpha_i = 0 \quad \forall i = 0, \dots, n.$$

$\Rightarrow B$ est une famille libre, et $\text{card}(B) = n+1 = \dim(\mathbb{R}_n[X])$

Donc B est une base de $\mathbb{R}_n[X]$.

b) $S = \text{Pass}(B_C, B)$

$$= \begin{pmatrix} x^n & x^{n-1} & \dots & x & 1 \\ 0 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & 0 \\ 1 & 0 & \dots & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ x \\ x^2 \\ \vdots \\ x^n \end{pmatrix}$$

Correction exercice 6: Serie Calculs matriciels

1. $U(P) = -2P + (x+1)P'$

Soit $A, B \in \mathbb{R}_n[x]$

$\alpha \in \mathbb{R}$

$$\begin{aligned} U(\alpha A + B) &= -2(\alpha A + B) + (x+1)(\alpha A + B)' \\ &= -2\alpha A + (x+1)\alpha A' - 2B + (x+1)B' \\ &= \alpha(-2A + (x+1)A') - 2B + (x+1)B' \\ &= \alpha U(A) + U(B) \end{aligned}$$

$\Rightarrow U$ est une application Pincorne

Soit $P \in \mathbb{R}_n[x] \Rightarrow \deg P \leq n$

$\deg(P') \leq n-1$

$\deg((x+1)P') \leq n-1+1 \leq n$

$\Rightarrow \deg(U(P)) \leq n$

donc $U(P) \in \mathbb{R}_n[x]$

d'où U est un endomorphisme de $\mathbb{R}_n[x]$

2) $A = \text{mat}(U, B \in) =$

$$\begin{pmatrix} U_1 & U(x) & \dots & U(x^k) & \dots & U(x^m) \\ -1 & 1 & \dots & 0 & \dots & 0 \\ 0 & -1 & \dots & \vdots & \dots & \vdots \\ \vdots & \vdots & \ddots & k & \dots & 1 \\ 0 & 0 & \dots & k-2 & \dots & m \\ 0 & 0 & \dots & 0 & \dots & m-1 \\ 0 & 0 & \dots & 0 & \dots & 0 \end{pmatrix} \begin{pmatrix} 1 \\ x \\ \vdots \\ x^k \\ \vdots \\ x^m \end{pmatrix}$$

$U(1) = -2$

$U(x) = -2x + (x+1) = -x+1$

$U(x^k) = -2x^k + kx^k + kx^{k-1} = (k-2)x^k + kx^{k-1}$

$U(x^m) = (m-2)x^m + m(x)^{m-1}$

$$3) a) P_k = X^{n-k}, \quad 0 \leq k \leq n$$

$$B = (P_k; 0 \leq k \leq n)$$

$$\left. \begin{array}{l} P_0 = X^n \\ P_1 = X^{n-1} \\ \vdots \\ P_{n-1} = X \\ P_n = 1 \end{array} \right\} \text{base canonique} \\ \text{de } \mathbb{R}_n[X] \text{ inversée}$$

on $(1, X, \dots, X^n)$ est la Base de $\mathbb{R}_n[X]$

alors cette famille est p.f.b.e

$$\text{Soient } \alpha_0 \dots \alpha_n \in \mathbb{R} / \sum_{i=0}^n \alpha_i P_i = 0$$

$$\Leftrightarrow \alpha_0 P_0 + \alpha_1 P_1 + \dots + \alpha_n P_n = 0$$

$$\Leftrightarrow \alpha_0 X^n + \alpha_1 X^{n-1} + \dots + \alpha_{n-1} X + \alpha_n = 0$$

$$\Leftrightarrow \alpha_i = 0 \quad \forall i = 0, \dots, n$$

$$\Rightarrow B \text{ est une famille p.f.b.e et } \text{card}(B) = n+1 = \dim \mathbb{R}_n[X]$$

Donc B est une Base de $\mathbb{R}_n[X]$

$$b) S = \text{Mat}(B_C, B)$$

$$= \begin{array}{c} \begin{array}{cccc} & X^n & X^{n-1} & \dots & X & 1 \end{array} \\ \left(\begin{array}{cccc} 0 & 0 & \vdots & 0 & 1 & 0 \\ 0 & 0 & \vdots & 0 & 0 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 1 & \vdots & 0 & 0 & 0 \\ 1 & 0 & \vdots & 0 & 0 & 0 \end{array} \right) \begin{array}{c} 1 \\ X \\ \vdots \\ X^{n-1} \\ X^n \end{array} \end{array}$$

$$c) M = \text{mat}_B(u)$$

$$= \begin{pmatrix} u(x^m) & u(x^{m-1}) & \dots & u(x) \\ m-2 & 0 & & 0 \\ m & m-3 & & 0 \\ 0 & m-1 & & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & & -2 \end{pmatrix} \begin{matrix} x^m \\ x^{m-1} \\ \vdots \\ 1 \end{matrix}$$

$$u(x^m) = -2x^m + (x+1)^m x^{m-1} \\ = (m-2)x^m + m(x^{m-1})$$

$$= (m-2)P_0 + mP_1$$

$$u(x^{m-1}) = (m-3)P_1 + (m-1)P_2$$

$$u(x) = -P_{m-1} + P_m$$

$$u(1) = -2P_m$$

$$d) M = \text{mat}(u, B)$$

$$A = \text{mat}(u, B_c)$$

$$S = \text{Pon}(B_c, B)$$

$$\left\{ \begin{array}{l} M = S^{-1} A \cdot S \end{array} \right.$$